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Fabry-Pérot interferometry for long range displacement sensing

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We investigate different optical configurations of a low-finesse Fabry-Pérot interferometer used for displacement sensing. The different configurations of the Fabry-Pérot cavity are selected in order to achieve large measurement ranges and angular alignment tolerances and to make the interferometer applicable for targets of various reflectivity ranges. The possible working ranges and angular alignment tolerances are characterized with respect to the interference contrast which is a measure for the signal quality. The use of a confocal arrangement enables a measurement range of up to about 0.4 m, or to work with an angular tolerance of more than $\pm 0.2^\circ$. In order to predict the optical response of arbitrary configurations of the Fabry-Pérot interferometer, we introduce a simulation method based on the Airy formula and the fiber optic coupling efficiency. © 2013 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4821623>]

I. INTRODUCTION

Length metrology with nanometer precision and beyond plays a decisive role in a variety of today's industrial and scientific applications. A metrology tool suited to measure displacements has to meet different requirements with respect to robustness, usability, size, working range, and precision. In integrated circuit manufacturing, for example, stage positioning and wafer alignment with nanometer precision over measurement ranges of up to several hundreds of millimeters is necessary to improve overlay in double patterning lithographic processes.^{1,2} Future semiconductor production will be based on extreme ultraviolet lithography which further requires the metrology system to be operated in a vacuum. In scanning force microscopy, an accurate spatial resolution can only be obtained when the position of the probe is tracked during surface scanning, which is typically in the micrometer range.³ Since operation under extreme environments, such as cryogenic temperatures, ultra high vacuum, or high magnetic fields, are highly desired, this technique sets especially high requirements for the robustness and the size of the metrological system. The sensor must fit into a cryostat with very constrained space and tolerate deformations caused by large temperature gradients which might lead to misalignment of the system. In particle accelerators, a large number of devices, such as steering and focusing magnets, have to be aligned with respect to the particle beam with nanometer precision,^{4,5} thus driving the need for low-cost multi-axis systems.

In this work, we present an interferometric system that can meet the requirements described above. It combines all advantages of a fiber-optic Fabry-Pérot interferometer with different cavity designs allowing to achieve a large measurement range with large angular alignment tolerance and to use different reflector materials. A large angular alignment tolerance allows for deformations of the sensing cavity or lat-

eral displacements of a moving stage to a certain extent and makes the interferometer easy to align. The system is based on telecom wavelength (1550 nm) components and optical fibers. This enables the miniaturization of the whole system and reduces its costs. The use of optical fibers makes the interferometer robust against environmental influences and allows remote sensing, thus promoting noninvasive operation.⁶ The Fabry-Pérot interferometer further supports the robustness and the miniaturization of the system as it includes no reference arm and no beam splitters. These advantages are especially desirable for displacement sensors that should be operated at cryogenic temperatures or at high magnetic fields and achieve sub-nanometer resolution and accuracy, conditions where most known commercial interferometers would fail.

The Fabry-Pérot interferometer can be based on either a low-finesse or a high-finesse cavity, formed between the end-face of a cleaved optical fiber and the reflecting surface of a movable target reflector. An example for the successful use of a high-finesse cavity for displacement sensing is the work of Lawall⁷ and Durand *et al.*,⁸ who reported an uncertainty of 10 pm and a sub-pm uncertainty, respectively, but with the drawback of increased system complexity. Less complex systems can be realized using a low-finesse Fabry-Pérot interferometer, which we will focus on in this work. A successful implementation is presented in the work of Smith *et al.*,⁹ who reported a sub-pm resolution even for DC displacement measurements. In our present work, different configurations of the Fabry-Pérot cavity are investigated and characterized with respect to their ability to increase the measurement range of the interferometer and to their tolerance against angular misalignment. We analyzed a number of representative practical configurations. In order to investigate the interference signal for a large parameter range, we performed simulations tested to reproduce the experimental results. The simulations can help to simplify the development process and reduce the testing time for different configurations.

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II. INSTRUMENTATION

The fiber-optic Fabry-Pérot interferometer setup is schematically shown in Fig. 1 with three different sensor head arrangements (a)–(c). The light source is a tunable distributed feedback (DFB) laser with a wavelength of $\lambda = 1535$ nm. The laser source is connected to one of the two input arms of a two by two directional fiber coupler which splits the light at a ratio of 50% on each arm. One of the output arms is connected to a fiber patch cable. The polished fiber end forms the first semi-transparent mirror of the Fabry-Pérot cavity and provides the reference beam for the interferometer as it reflects about 4% of the incoming light. 96% of the light is coupled into the cavity and reflected off a reflector attached to a movable target. The reflected light is coupled back into the fiber and interferes with the reference beam, thus producing an interference pattern which is detected at the second input arm of the coupler using an InGaAs detector. Since we were only interested in the maximum and minimum values of the reflected signal, the laser wavelength was swept at a fixed target position to go through the interference pattern and only the intensities at the fringe extrema were recorded.

An example for the detected interference fringes is shown in Fig. 2. It was generated by applying a thermal drift to a cavity placed in a vacuum. The data were recorded with a bandwidth of 1 kHz. The root mean square (rms) noise at quadrature corresponds to an equivalent displacement of 0.04 nm.¹⁰

The different Fabry-Pérot sensor head configurations used for displacement sensing are schematically shown in Fig. 1(a)–1(c). Configuration (a) is a lens-less system where the beam directly impinges on the target. In (b), the light emitted by the fiber is collimated by a lens and in (c), the beam is focused by the same lens. For characterization of measurement range and angle tolerance, both constructive and destructive interference intensities are recorded as a function of target tilt angle α and working distance WD . The two arrangements including a lens were realized with the help of an adjustable sensor head composed of two parts. The first part includes a connector for the FC/PC fiber and the second part carries the lens. Both parts are connected to each other by a screw to retain a constant fiber to lens separation. In this

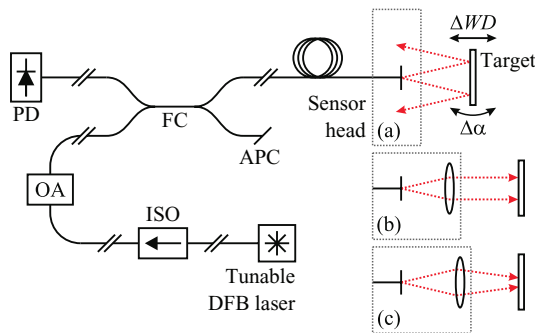


FIG. 1. Fiber-optic measurement setup including laser source, isolator (ISO), optical attenuator (OA), fiber coupler (FC), photodetector (PD), and optical Fabry-Pérot cavity, formed between fiber end and target. All parts are interconnected by FC/APC to FC/APC mating sleeves. Shown are also three different configurations of the sensing head, comprising (a) only fiber and target; (b) fiber, collimating lens, and target; and (c) fiber, focusing lens, and target.

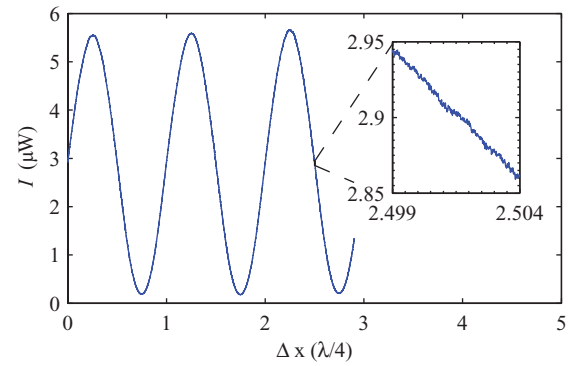


FIG. 2. Interference pattern produced by sweeping the temperature of a 30 mm long cavity located inside a vacuum chamber. The inset shows the signal around the maximum slope.

way, different beam geometries can be easily realized. In this work, a lens with a focal length $f \approx 8$ mm and a numerical aperture $NA = 0.50$ was used. For the tests, the sensor head is mounted on a travel stage in order to change the cavity length, whereas the target tilts are realized through a gimbal center mirror mount including piezo-assisted micrometer drives.

III. REFLECTIVITY OF THE FIBER COUPLED FABRY-PÉROT CAVITY

As already indicated in Fig. 1, different configurations of the sensor head are evaluated, which are applied depending on the required specifications. For theoretical characterization and customization of the sensor heads discussed in this work, we developed a model to predict and test the optical behavior of different sensor head designs with respect to interferometric signal quality and tolerances. This considerably eases the development and manufacturing process of new sensor head designs and reduces the testing time. The model is based on the Airy formula¹¹ and describes the reflectivity of a Fabry-Pérot interferometer. It takes into account different reflectivities of the cavity surfaces as well as the fiber coupling efficiency of the system, which denotes the percentage of power transmitted into the fiber with respect to the power emitted from the fiber.^{12,13} In this way, the interference at the fiber endface can be determined analytically. The cavity, which behaves like an optical resonator, is formed between the cleaved endface of an optical single mode fiber with reflectivity R_1 and a tiltable target with reflectivity R_2 . The linear polarized laser light emitted from the laser is expressed by the electromagnetic wave $E = E_0 e^{i\omega t}$. It is routed to the fiber endface, where part of the light is reflected by the glass/air interface with the amplitude reflection coefficient r_1 , thus forming the reference beam. The remaining part exits the fiber with the amplitude transmission coefficient t_1 under a specific cone angle which is defined by the numerical aperture (NA) of the fiber. In the cavity, the light can take different paths depending on the used lenses and the target tilt. Along this path, the light resonates between the Fabry-Pérot mirrors with the amplitude reflection coefficients r_1 and r_2 , given by $r_1 t_1^* = R_1$ and $r_2 t_2^* = R_2$. After each cycle, part of the light is transmitted into the fiber with the amplitude transmission coefficient

t'_1 where it interferes with the reference beam. Since the light transmission into the fiber also depends on the angle of incidence, which can vary with the target tilt, our model also has to consider the power-coupling efficiency η of the fiber. When taking the square root of η , we can apply the power-coupling efficiency to electromagnetic fields using the relation $t'_1 = \sqrt{\eta}t_1$. Summing over all portions which are coupled back into the fiber yields an expression for the reflected electromagnetic wave E_r . The product $E_r \cdot E_r^*$ finally is proportional to the reflected intensity I_r ,¹¹ according to

$$I_r = I_0 \frac{r_1^2 + (fr_2)^2 - 2r_1fr_2 \cos \varphi}{1 + (r_1r_2)^2 - 2r_1r_2 \cos \varphi}, \quad (1)$$

where

$$f = r_1^2 + \sqrt{\eta}t_1^2. \quad (2)$$

Equation (1) takes already into account the phase shift which occurs between reflections within the cavity. The amplitude transmission coefficients t_1 and t'_1 for the waves traveling from the fiber to the cavity and vice versa, respectively, are related to the reflectivity R_1 and the transmissivity T_1 of the glass/air interface by $t_1t_1^* = T_1 = 1 - R_1$. The optical phase difference φ between two consecutive interfering waves is given by

$$\varphi = \frac{4p\pi nz}{\lambda}, \quad (3)$$

where z is the cavity length, n is the refractive index of the medium in the cavity, and p is the folding order of the cavity, which indicates whether the beam passes the cavity only once ($p = 1$, single pass) or twice ($p = 2$, double pass).

The quality of the interferometric signal is quantified by the fringe contrast or visibility V and by the finesse of the cavity $\mathcal{F}$. The finesse of the cavity is given by¹⁴

$$\mathcal{F} = \frac{\pi \sqrt{r_1 r_{2,\text{eff}}}}{1 - r_1 r_{2,\text{eff}}}, \quad (4)$$

where we introduced the concept of an effective reflection coefficient $r_{2,\text{eff}}$, given by $r_{2,\text{eff}} = fr_2$. The cavity finesse can be seen as a measure for how close the interference pattern is to a sinusoidal shape. As the measurement range of the interferometer is intended to exceed the linear region of the interference pattern, the interferometer has to be able to track phase changes over a full period of 2π with constant sensitivity and high linearity. This can be achieved, for example, by applying a quadrature detection method based on the phase generated carrier approach.^{15,16} This method requires a sinusoidal interference pattern and a 90° phase shifted signal, generated by demodulation of the carrier signal. The sinusoidal shape of the interference pattern can be ensured by a low-finesse of the sensing cavity.

The fringe contrast is given by

$$V = \frac{I_{\max} - I_{\min}}{I_{\max}}, \quad (5)$$

where $I_{\max}$ and $I_{\min}$ are the intensities for constructive and destructive interference, respectively. The contrast is of particular importance because it is proportional to the signal-to-noise ratio of the detected signal and thus to the sensitivity of

the displacement sensing.⁹ In this work, we will focus on the contrast and consistently only show data of the interference extrema, corresponding to constructive and destructive interference. These intensities can be calculated by inserting the coupling efficiency η into Eq. (1) and calculating $I_r(\varphi = \pi) = I_{\max}$ for constructive and $I_r(\varphi = 0) = I_{\min}$ for destructive interference. Eq. (1) shows that the contrast crucially depends on the reflectivity of the movable target. This is demonstrated in Fig. 3, which shows constructive and destructive interference intensities and the corresponding contrast as a function of the target reflectivity R_2 , assuming $\eta = 1$. The contrast is maximum around a target reflectivity that equals the fiber core, which is $R_2 = 0.0359$. For higher reflectivities, the contrast decreases because of the increasing amount of light coupled back into the fiber. In this case, the contrast could be increased by reducing the coupling efficiency, since $r_{2,\text{eff}}$ also depends on the coupling efficiency according to Eq. (1). High target reflectivities thus can generally be compensated by reducing the coupling efficiency in order to achieve a higher contrast.

IV. COUPLING EFFICIENCY SIMULATION

The amount of light coupled back into the source fiber is described by the coupling efficiency η , which is calculated as a function of target tilt and working distance using an optics simulation software. When performing fiber coupling simulations, the NA of the optical fiber is required. Unfortunately, the definition of NA is not standardized. The manufacturers of optical fibers define it as the sine of the half light cone angle at which the far field intensity of the light exiting the fiber has fallen to 1% of its peak value ($\text{NA}_{1\%}$), while in laser optics it is commonly defined by the $1/e^2$ power level (NA_{1/e^2}). Optical software providers, such as Zemax,¹⁷ however, use two different definitions within their own software. In the following, we will refer to Zemax and provide a useful correspondence between all definitions.

In some places, the Zemax fiber coupling feature requires a source and receiver fiber NA which is defined by NA_{1/e^2} . The Corning data sheet¹⁸ provides for the fiber we used $\text{NA}_{1\%} = 0.14$, which can be converted to the $1/e^2$ power level using $\text{NA}_{1/e^2} = \text{NA}_{1\%} \cdot (2/\ln 100)^{1/2}$. It follows

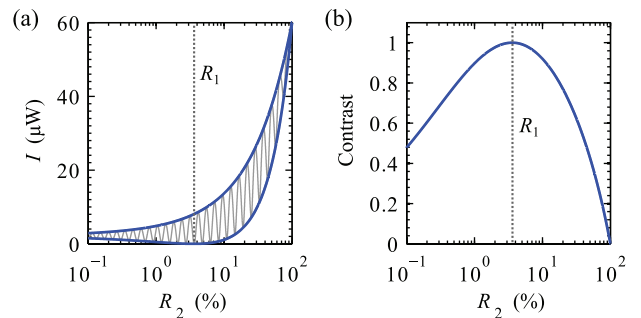


FIG. 3. Calculated constructive (upper curve) and destructive (lower curve) interference intensities I (a) and corresponding contrast (b) as a function of the target reflectivity R_2 . Maximum contrast is achieved for equal target and fiber core reflectivities (here about 3.6%). The interference pattern in between is just for schematic illustration.

$NA_{1/e^2} = 0.092$ for our fiber. In other places in the same software, the “object space numerical aperture,” which refers to the source fiber NA, is twice the sine of the paraxial marginal ray angle, i.e., the full cone angle diverging from the source fiber, also called acceptance angle. It is calculated from the refractive index of core and cladding, resulting in $NA = 0.25$ according to $NA = (n_{\text{core}}^2 - n_{\text{clad}}^2)^{1/2}$, where $n_{\text{core}} = 1.4682$ is the index of refraction of the fiber core and $n_{\text{clad}} = 1.4629$ is the index of refraction of the fiber cladding at a wavelength of $\lambda = 1550$ nm according to the Corning data sheet.

V. POSITION SENSING FOR LENS-LESS CAVITIES

The sensing cavity can take different shapes depending on the specific application. In a first configuration of the Fabry-Pérot cavity, the light emitted from the fiber is directly reflected by the target and coupled back into the fiber, as shown in Fig. 4(a). This arrangement is especially suited for applications with very constrained space. It can be found in atomic force microscopes for measuring the cantilever deflection¹⁰ or in piezoelectric scanners for expansion measurements.

An example for this interferometer type is shown in the work of Smith *et al.*,⁹ where a glass wedge forms the movable reflector in order to get equal reflectivities of both cavity surfaces. Such a balanced low-finesse cavity is crucial to achieve a high signal-to-noise ratio. Due to the divergence of the light emitted from the fiber, this system is limited to cavity lengths not exceeding $5 \mu\text{m}$, a range which is too small for most applications. Extending the measurement range of this type of interferometer and at the same time preserving its large angular alignment tolerance will be the central goal of this work.

A good signal contrast can be achieved by either using a glass target placed in close vicinity to the fiber or by using a high reflective target placed far enough away so that the beam divergence reduces the back coupled power to approximately 4% of the total power. To determine the position of maximum contrast for high reflective targets, the position dependent power coupling efficiency must be known. For normal incidence, it can be analytically calculated by the convolution of two Gaussian beam profiles, one representing the field distribution of the acceptance fiber Ψ_1 and the other one representing the field distribution Ψ_2 after a distance of two times the cavity length z ,

$$\eta = \left| \int d^2r \Psi_1 \Psi_2^* \right|^2. \quad (6)$$

The radial field distribution Ψ at a distance ζ from the source fiber is given by

$$\Psi(\zeta) = \sqrt{\frac{2}{\pi}} \frac{1}{w(\zeta)} \exp \left\{ - \left(\frac{r}{w(\zeta)} \right)^2 \right\}, \quad (7)$$

where w is the beam waist size, and r is the radius with respect to the fiber optical axis. The normalized power coupling efficiency η is then given by the convolution of the field distributions at a distance $\zeta_1 = 0$ and $\zeta_2 = 2z$,

$$\eta = \left\{ \frac{2}{\pi w_1 w_2} \int_0^{2\pi} d\varphi \int_0^\infty dr \exp \left[- \left(\frac{1}{w_1^2} + \frac{1}{w_2^2} \right) r^2 \right] \right\}^2 = \left[\frac{2w_1 w_2}{w_1^2 + w_2^2} \right]^2. \quad (8)$$

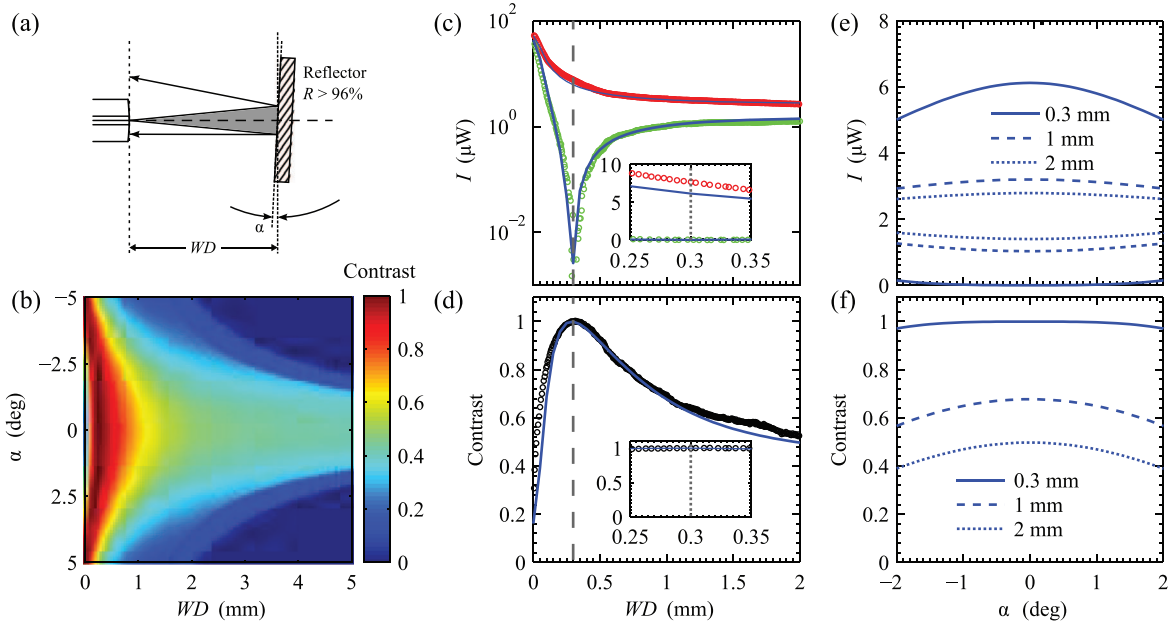


FIG. 4. Angle and position dependence for the configuration shown in (a), consisting of a mirror that directly reflects the light emitted from the fiber. (b) Simulated contrast as a function of working distance WD and target tilt angle α . (c) Position dependence of the constructive and destructive interference intensities I at normal incidence ($\alpha = 0$). Red and green circles denote measurement values for constructive and destructive interference, respectively. Blue lines represent simulation data. (d) Corresponding contrast. Black circles represent measurement values for the contrast. (e) Angle dependence at $WD = 0.3$ mm (position of maximum contrast), $WD = 1$ mm, and $WD = 2$ mm. (f) Corresponding contrast.

The beam waist at a distance ζ from the fiber is

$$w(\zeta) = w_1 \sqrt{1 + (\zeta/z_R)^2}, \quad (9)$$

where z_R is the Rayleigh length, given by

$$z_R = \frac{\pi w_1^2}{\lambda}. \quad (10)$$

Inserting Eqs. (9) into (8) yields the coupling efficiency as a function of the target distance z for normal incidence, which can be used to calculate the interference intensities and the contrast using Eq. (1),

$$\eta(z) = \left\{ \frac{2\sqrt{1 + (2z/z_R)^2}}{2 + (2z/z_R)^2} \right\}^2. \quad (11)$$

The resulting contrast is shown by the calculated line in Fig. 4(d) and is identical to the results from the simulations.

Apart from this analytical approach, simulations have been carried out to quantify the signal change for target tilts with respect to the fiber optical axis and for target movements along this axis. The results are shown in Figs. 4(b)–4(f) and compared to position dependent measurements at normal incidence. For small fiber to target distances, the signal intensities are in the order of I_0 and decrease very fast with increasing working distance, whereas the contrast becomes maximum when the returning light intensity approaches 4%. For normal incidence, the maximum contrast is reached at a distance of 0.3 mm from the fiber (cf. Fig. 4(d)). The inset of Fig. 4(c) shows the signals in a range of 50 μm around the maximum contrast. This area is well suited for piezo scanner position readout as the signal contrast is well above 0.98 in this range.

VI. TOWARDS INCREASING THE MEASUREMENT RANGE

One way to increase the measurement range of a low-finesse Fabry-Pérot interferometer is to put a collimating lens into the beam path between the fiber and glass target, as shown in Fig. 5(a). When the light hits the reflector at normal incidence and the reflector is made of glass with a reflectivity similar to the fiber core, the signal contrast becomes maximum as demonstrated in Figs. 5(b) and 5(c). The figures show both constructive and destructive interference intensities and the corresponding contrast as a function of the target tilt angle α . Both interference intensities are approaching $I_{r1} = 2.1 \mu\text{W}$ for large angles, meaning that the coupling of light from the cavity into the fiber nears zero, so only the reflection from the fiber end I_{r1} contributes to the detected intensity. The total intensity I_0 which is routed to the cavity is then $I_0 = I_{r1}/R_1 = 52.5 \mu\text{W}$, where $R_1 = 0.0359$ is the reflectivity of the fiber core. The data reveal the main problem of this approach, namely, a very high sensitivity of the interference signal contrast to target angular misalignments. This makes this arrangement unsuitable for most displacement tracking applications.

A further arrangement might use a retro reflector instead of a plane reflector to increase the angle tolerance of the displacement measurement. To determine the angle tolerance, an aluminum coated corner cube retro reflector was mounted on

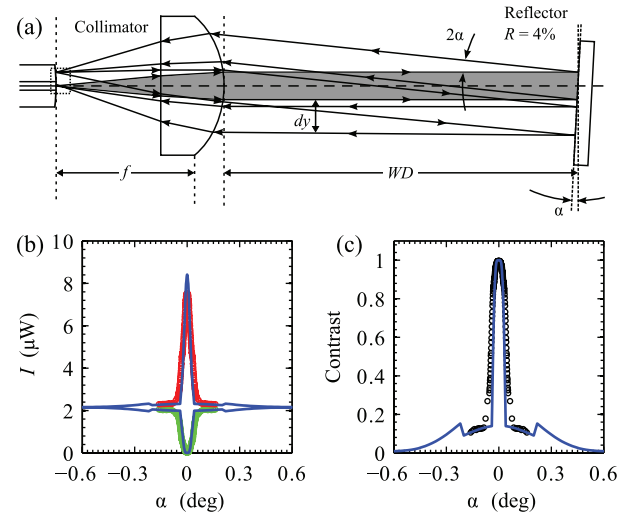


FIG. 5. Angle dependence for the configuration shown in (a), consisting of a lens which collimates the light emitted from a fiber and a plane glass target that reflects part of the light at a working distance of 31 mm. Red and green circles denote measurement values for constructive and destructive interference, respectively. Blue lines represent simulation data. Black circles represent measurement values for the contrast.

a tilt stage at a working distance of $WD = 400 \text{ mm}$, as depicted in Fig. 6(a). When increasing the target angle, i.e., the angle between the retro reflector optical axis and the incident beam optical axis, the interference signals decrease as shown in Fig. 6(b), whereas the contrast increases according to Fig. 6(c). The angle dependent decrease of the interference intensities result from a reduction of the light that is coupled back into the source fiber, caused by an offset between the fiber optical axis and the reflected beam. At 0° , the light emitted into the cavity is expected to be completely coupled back into the fiber, thus causing the interference intensities to be around I_0 . As this is not the case, there must be imperfections

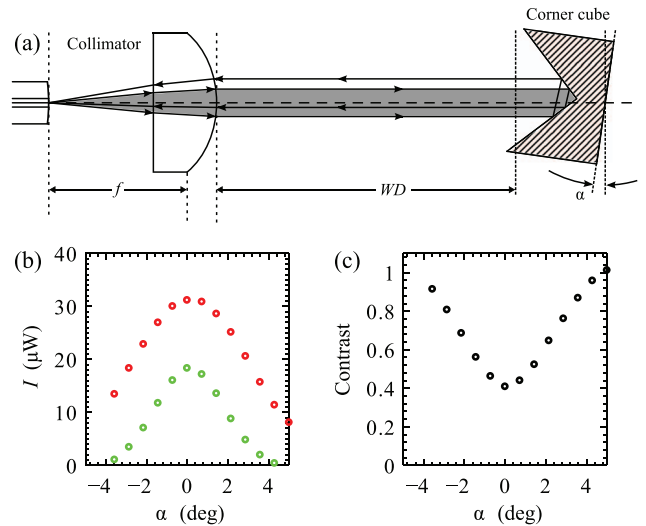


FIG. 6. Measured angle dependence for the configuration shown in (a), consisting of a lens which collimates the light emitted from a fiber and an aluminum coated corner cube retro reflector placed at a working distance of 400 mm. Red and green circles denote measurement values for constructive and destructive interference, respectively. Black circles represent measurement values for the contrast.

in the corner cube which prevent the light from being reflected parallel to the incident beam, thus causing losses which reduce the power at the detector. For practical applications, the contrast can be further improved by tilting or decentering the corner cube.

VII. POSITION SENSING USING A “SELF-ALIGNING SENSOR HEAD”

Section VI shows that a tedious alignment of the reflector is required to couple the light back into the fiber when using a low reflective plane target, thus making this technique hard to use and sensitive to misalignments like lateral displacements which might occur during target movement. A corner cube retro reflector might relax the alignment procedure but for some applications it is too bulky and more expensive and it can suffer from a weak signal contrast in case of a high reflectivity.

A new approach uses a high reflective plane target slightly tilted with respect to the incident collimated beam to form a confocal arrangement as shown in Fig. 7(a). In this configuration, the beam is not directly coupled back into the fiber. It rather makes use of the small target misalignment causing the reflected light to be focused on the fiber cladding or the ferrule of the connector surface.¹⁹ Depending on the

reflectivity of cladding or ferrule, a certain amount of light is reflected off the connector surface and passes the cavity a second time. Since the returning beam is parallel to the incident collimated beam, it will be exactly focused to the fiber core after having passed the cavity twice. This means that the cavity has a kind of self-alignment mechanism that makes the coupling of the reflected light into the fiber independent from the target alignment. At the same time, the balanced reflectivities of the fiber core and cladding allow to achieve a high signal contrast of the interference pattern. Additionally, their low reflectivities, which are in the order of 4%, make the cavity low-finesse, resulting in a more sinusoidal interference pattern which again increases the accuracy of long range displacement measurements.

To demonstrate the signal tolerance against target tilts and displacements for a wide parameter range, simulations have been carried out, for which the arrangement is investigated in greater detail. The optical system in Fig. 7(a) can be described as a folded Fabry-Pérot cavity which is set up between the fiber core and cladding, respectively, ferrule, separated by a cavity length which equals about twice the distance between fiber end and target. The reflector only serves as a folding element which has no influence on the finesse of the cavity. The crucial point of this technique is the beam focusing to the fiber connector surface. For this reason, we will

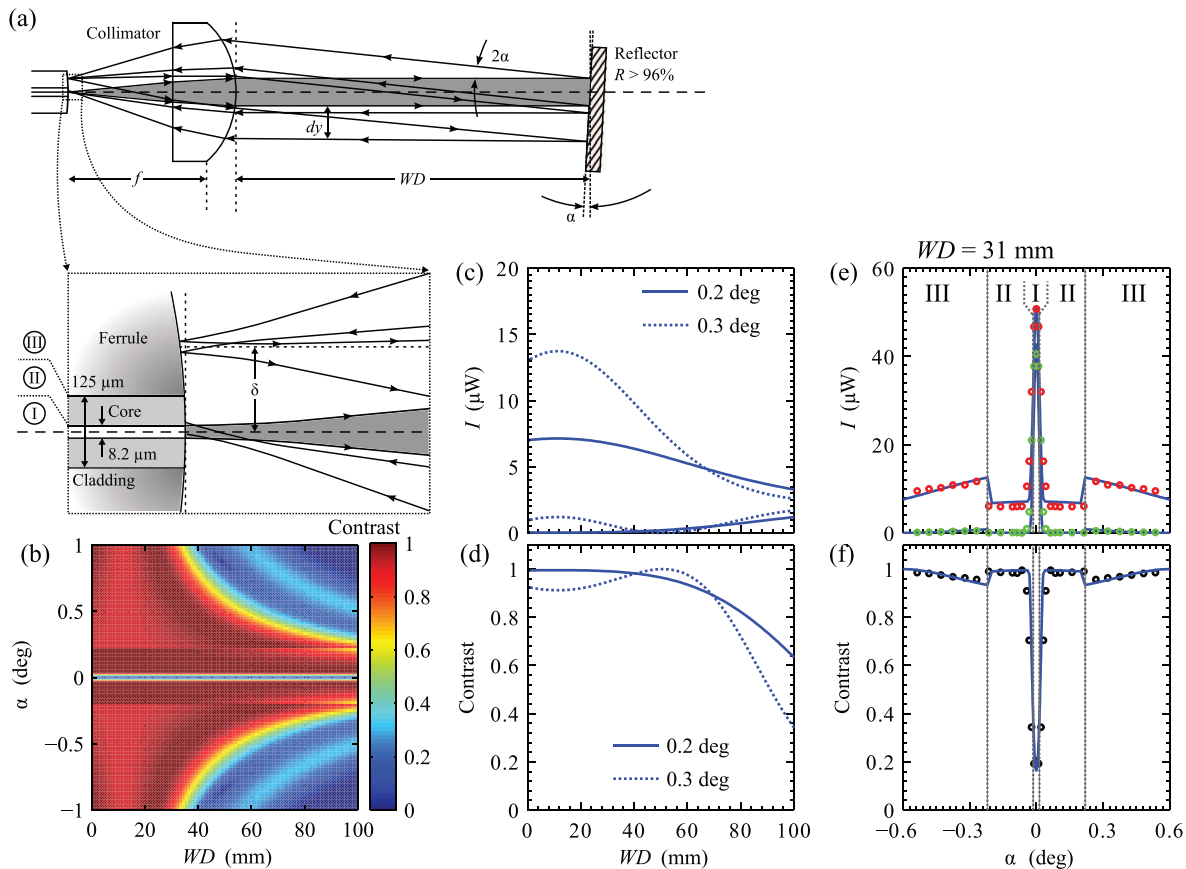


FIG. 7. Angle and position dependence for the configuration shown in (a), representing a confocal system consisting of a high reflective plane mirror and a lens that collimates the light emitted from the fiber. (b) Simulated contrast as a function of working distance WD and target tilt angle α . (c) Position dependence at $\alpha = 0.2^\circ$ and $\alpha = 0.3^\circ$. Red and green circles denote measurement values for constructive and destructive interference, respectively. Blue lines represent simulation data. (d) Corresponding contrast. Black circles represent measurement values for the contrast. (e) and (f) Angle dependence at $WD = 31$ mm. Points denote measured data and lines denote results from the model calculation.

focus on the beam geometry near the fiber, which is shown in more detail in the inset of Fig. 7(a). For the given configuration, the angle between incident beam and its first reflection is twice the tilt angle α of the reflector. It follows that the distance δ between the center of the fiber core and the reflected spot is

$$\delta = f \tan(2\alpha), \quad (12)$$

where f is the effective focal length of the collimator lens. The reflected spot consequently moves along the radial direction on the connector surface when the reflector surface normal is tilted away from the fiber optical axis, thus relaxing the alignment procedure. The endface of an optical single mode FC/PC connector is composed of three parts, namely, the fiber core, the cladding, and the ferrule. The fiber core has a diameter of $8.2 \mu\text{m}$ and an index of refraction of $n_{\text{core}} = 1.4682$ at a wavelength of $\lambda = 1550 \text{ nm}$,¹⁸ leading to a reflectivity of $R_{\text{core}} = 0.0359$ in standard air with an index of refraction of $n_{\text{amb}} = 1.0003$, according to

$$R_{\text{core}} = \left| \frac{n_{\text{core}} - n_{\text{amb}}}{n_{\text{core}} + n_{\text{amb}}} \right|^2. \quad (13)$$

The cladding has a diameter of $125 \mu\text{m}$ and an index of refraction of $n_{\text{clad}} = 1.4629$ at the same wavelength, yielding a reflectivity of $R_{\text{clad}} = 0.0353$. The ferrule is made of zirconia and has an index of refraction of $n_{\text{ferrule}} = 2.1252$ at $\lambda = 1550 \text{ nm}$, yielding a reflectivity of $R_{\text{ferrule}} = 0.1295$ in air. Usually, the endface of a FC/PC connector is not flat. It has a radius of curvature of about $r_{\text{ferrule}} = 20 \text{ mm}$, which also has to be considered in the simulations. To simulate the whole complexity of the folded cavity with respect to interference intensities, the simulations were subdivided into three cases, characterized by the radial position δ of the reflected spot on the fiber surface or by the tilt angle α , which is related to δ by Eq. (12). For small angles, referring to $|\delta| < 4.1 \mu\text{m}$ or $|\alpha| < 0.014^\circ$ (regime I), the Fabry-Pérot cavity is set up between fiber end and mirror and the beam passes the cavity only once (single pass), as the spot lies on the fiber core. This means that the reflectivities of the target and the core have to be considered in the transfer function calculations of the cavity. When the tilt is further increased, corresponding to $4.1 \mu\text{m} < |\delta| < 62.5 \mu\text{m}$ or $0.014^\circ < |\alpha| < 0.220^\circ$ (regime II), the configuration enters the regime in which the beam passes the cavity twice (double pass). In this regime, the beam is focused on and reflected off the fiber cladding and the mirror only has a folding function. This means that the Fabry-Pérot cavity is set up between the fiber core and cladding, leading to a cavity length that is about twice the distance between fiber and target and the folding order is $p = 2$. For tilts exceeding 0.220° or $|\delta| > 62.5 \mu\text{m}$ (regime III), the configuration is still in double pass mode, but the beam is focused on the ferrule, leading to a higher finesse of the cavity because of the higher reflectivity of the ferrule.

The simulation results are shown in Figs. 7(b)–7(f) and compared to angle dependent measurements taken at a working distance $WD = 31 \text{ mm}$ for verification. Fig. 7(b) shows simulations of the distance and angle dependent contrast. In the high contrast regimes II and III, the back coupled signal power decreases with both increasing working distance and

target tilt, which shows up in a decrease of contrast. This decline is caused by the limited acceptance angle of the fiber which reduces the amount of light that is coupled into the fiber, i.e., the coupling efficiency, the larger the angle of incidence towards the fiber is. An increasing tilt or working distance increases the distance between the incident and the last reflected beam (denoted as d_y in Fig. 7(a)), thus increasing the fiber incidence angle of the latter. The single pass regime I, however, has a weak contrast because of the large amount of light that is coupled back into the fiber and is, therefore, not suited for high resolution displacement interferometry, i.e., the target should always be slightly tilted. Fig. 7(c) shows both constructive and destructive interference intensities for working distances ranging from 0 to 100 mm at two exemplary angles that correspond to a reflection on the cladding ($\alpha = 0.2^\circ$, regime II) and the ferrule ($\alpha = 0.3^\circ$, regime III). Fig. 7(d) shows the corresponding contrast. The lower contrast for small distances at $\alpha = 0.3^\circ$ results from the higher reflectivity of the ferrule. When the distance to the target is further increased, the contrast becomes maximum as a result from a decreasing coupling efficiency which compensates the high reflectivity of the target. Angle dependent data are shown in Figs. 7(e) and 7(f) at a working distance of 31 mm. The simulations make a good fit to the experimental data. Deviations might arise from deviations of the ferrule curvature or from a badly collimated beam.

VIII. POSITION SENSING WITH FOCUSING OPTICS

Practice has shown that it is not always possible to attach a sufficiently large reflector to the displacing target. Applications further might require the measurement directly on rough surfaces, surfaces with low reflectivity or surfaces with curvatures. Other applications allow only a very small beam spot on a surface with arbitrary reflectivity. For all these applications, it is appropriate to use a focused beam which is easily achieved by moving the lens a bit further away from the back focal spot, as schematically shown in Fig. 8(a). Light being reflected off a target placed in the focus of the beam will always be refocused to the fiber core. This makes the configuration very tolerant against misalignment of the target and, thus, also applicable to rough surfaces. This configuration is especially well suited for targets with a low reflectivity, as the maximum contrast is achieved for a reflectivity of about 4%, assuming that $\eta = 1$. The small spot size at the focus point again promotes the use of rough surfaces and small targets. The $1/e^2$ intensity beam diameter at the focus point is twice the radial beam waist w , given by $w \approx \lambda/\pi\theta$, where θ is the half convergence angle of the beam propagating towards the target. High reflective targets, however, significantly decrease the fringe contrast when placed in the focus. Therefore, the coupling efficiency needs to be reduced, which can be achieved by defocusing the target. Since the technique for measuring displacements with a focused beam consequently depends on the target reflectivity, we distinguish in this section exemplarily between low ($R \approx 4\%$) and high reflective targets ($R > 96\%$) to illustrate the difference of both approaches. For both parts, the resulting signal and its sensitivity to target misalignment and working distance changes is

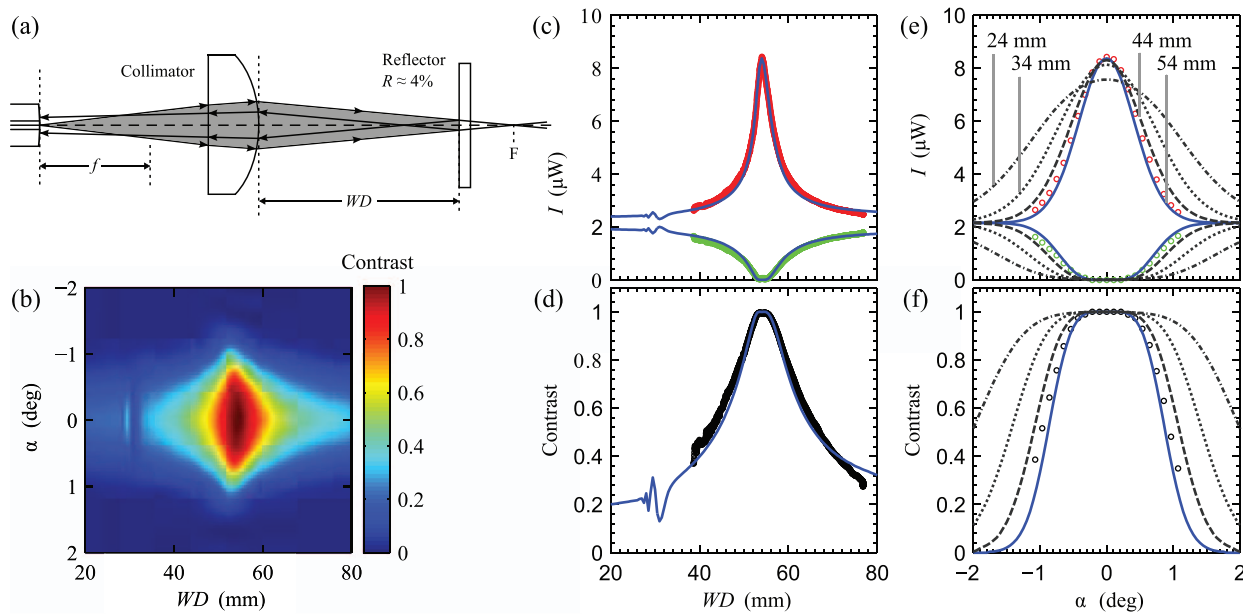


FIG. 8. Angle and position dependence for the configuration shown in (a), consisting of a lens that focuses the light from the fiber to a working distance of $WD = 54$ mm and a low reflective plane glass target. (b) Simulated contrast as a function of working distance WD and target tilt angle α . Red and green circles denote measurement values for constructive and destructive interference, respectively. Blue lines represent simulation data. (c) Position dependence of the constructive and destructive interference intensities I at normal incidence. (d) Corresponding contrast. Black circles represent measurement values for the contrast. (e) and (f) Angle dependence at the focus point with the beam focused to $WD = 24$ mm, $WD = 34$ mm, $WD = 44$ mm, and $WD = 54$ mm.

exemplary shown for one special arrangement of fiber, lens, and target. Here, the focusing lens is inserted into the beam path at a distance $l = 7.38$ mm from the fiber. This leads to a focus point at a working distance of $WD = 54$ mm with a beam width of $28 \mu m$. Basically, the focus point can be adjusted to different working distances by simply changing the fiber to lens distance in the sensor head. The ultimate accuracy of this method requires to take the Gouy phase shift into account, when the target shifts through the focus.²⁰

A. Target with low reflectivity

Displacement measurements of targets with low reflectivity are required in applications such as the positioning of telescope lenses with low reflectivity or the measurement of the expansion of a piezoelectric material with high surface roughness. In this case, the targets must be placed directly in the focus point of the lens in order to maximize the back coupled light intensity to ideally 4%, ensuring a large fringe contrast.

To demonstrate the signal tolerance against target tilts and displacements, measurements are carried out using a plane glass with a curved backside placed in the focus of the beam so that the plane surface forms the second semi-transparent mirror of the cavity. The curvature helps to reduce unwanted reflections from the backside. The lens is made from crown glass (BK7) with an index of refraction of $n_2 = 1.50065$, yielding a reflectivity of $R_2 = 0.040$ in dry air. The measurements are again compared to simulations covering a greater parameter range.

The analysis of the data shown in Fig. 8 reveals a good agreement between measurement and simulation. The reflectivity of the target, which is very similar to the fiber core,

leads to a very high contrast of $V = 0.996$ in the experiment and $V = 1.000$ in the simulation. The contrast drops to $V = 0.9$ for distances of about $\Delta WD = \pm 3$ mm around the focus point. In the same interval, the constructive signal drops by a factor of about 2, thus limiting the accuracy of large range measurements, as intensity changes are always interpreted as displacements. To ensure a measurement accuracy in the low nanometer regime, displacements should be limited to a few hundred nanometers around the focus point. Figs. 8(e) and 8(f) show the angle dependence at the focus point for different working distances of the focus point. For the focus point at $WD = 54$ mm, the contrast drops to $V = 0.5$ for tilt angles of $\alpha = \pm 0.9^\circ$ around the fiber optical axis, demonstrating the high tolerance of this method against target misalignment. The angle dependence can be further reduced by focusing the beam to smaller working distances, i.e., the system gets more tolerant to misalignment the smaller the distance to the focus is. The decrease of constructive interference intensity with decreasing focus distance at normal incidence originates from optical aberrations.

B. Target with high reflectivity

Other applications require the measurement of displacements of high reflective surfaces and, at the same time, only allow very small beam spots or curved surfaces. Examples are the measurement of cantilever movements²¹ or the measurement of the wobble of a rotating cylinder.²² Some applications, moreover, suffer from a limited space, as for example when working in cryostats that allow only very small sensor heads, but require a measurement range of several millimeters. A focusing sensor head with a small lens operated in a defocused mode can meet these demands. The slightly larger

target spot size enlarges also the reflected spot size, leading to a smaller coupling efficiency and, consequently, to a higher contrast. A large angle tolerance is still guaranteed because of the large area which is covered by the reflected beam on the fiber surface.

Fig. 9 shows both theoretical and experimental data for the system consisting of a focused beam and a high reflective mirror. According to the simulations in Fig. 9(b), the contrast has a minimum at the focus point due to the high coupling efficiency and the high target reflectivity, leading to a low displacement resolution at that point. According to Figs. 9(c) and 9(d), interferometry should be rather performed at working distances of about ± 5 mm around the maximum

contrast at $WD = 42$ mm and $WD = 65$ mm, yielding a contrast greater than $V = 0.9$. This working range is about twice the range measured with low reflective targets and the change of signal intensities is significantly reduced, thus promoting long range measurements. The points of maximum contrast are also very insensitive against misalignment, as shown in Figs. 9(e), 9(f) and 9(i), 9(j). According to these figures, a contrast of $V = 0.5$ is reached at a target tilt of $\Delta\alpha = \pm 0.8^\circ$. For large range measurements over several hundreds of nanometers, the intensity change has to be taken into account when accuracies of only a few nanometers are required. Figs. 9(g) and 9(h) additionally show the angle dependence at the focus point, illustrating again the good

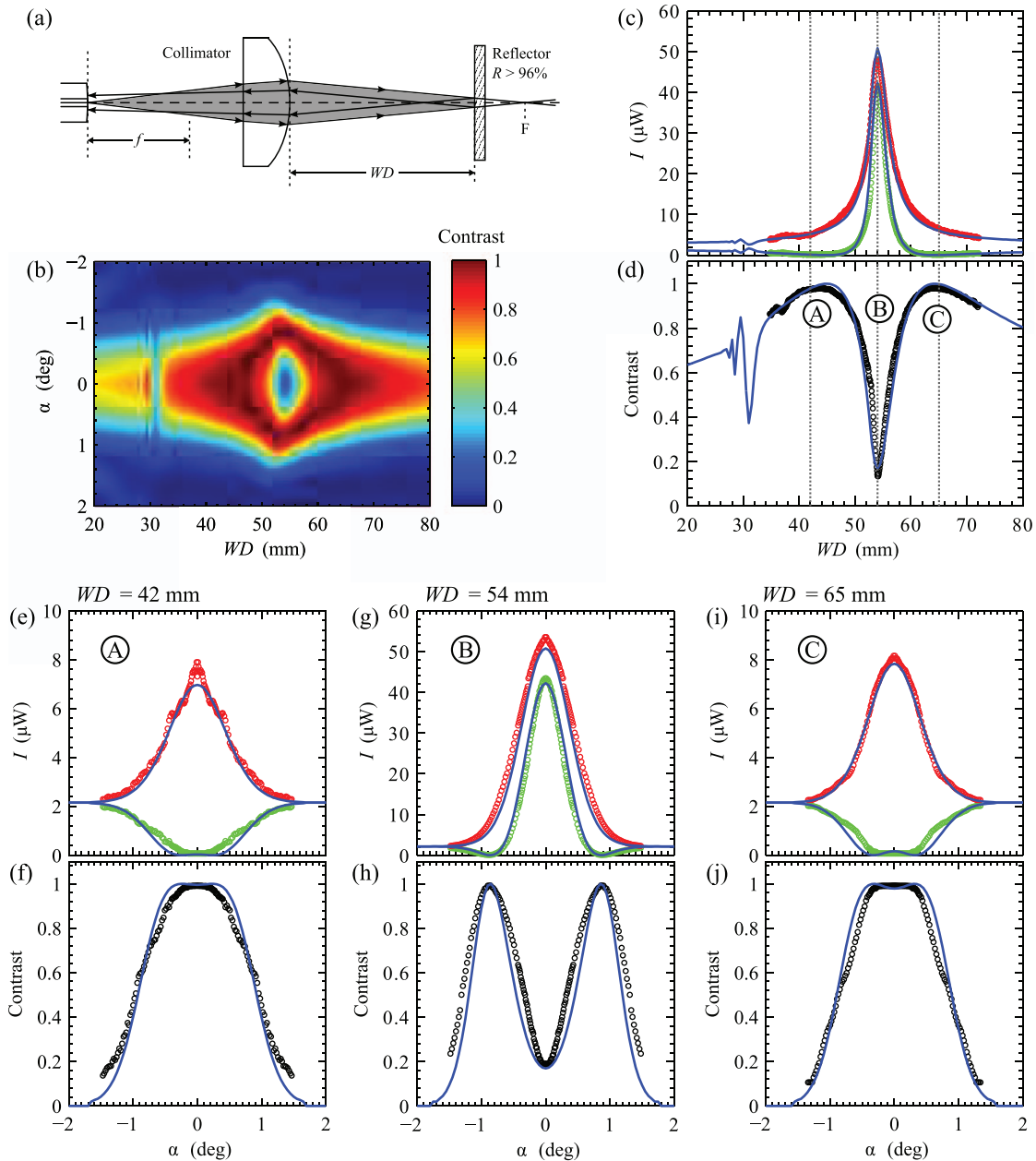


FIG. 9. Angle and position dependence for the configuration shown in (a), consisting of a high reflective plane mirror and a lens that focuses the light from the fiber to a working distance of $WD = 54$ mm. (b) Simulated contrast as a function of working distance WD and target tilt angle α . (c) Position dependence at normal incidence. Red and green circles denote measurement values for constructive and destructive interference, respectively. Blue lines represent simulation data. (d) Corresponding contrast. Black circles represent measurement values for the contrast. (e)–(j) Angle dependence at $WD = 42$ mm, $WD = 54$ mm (focus point), and $WD = 65$ mm. Points denote measured data and lines denote results from the model calculation.

accordance between measurements and simulations. The simulation results in Figs. 9(b)–9(d) further show oscillations in the intensities and the contrast around $WD = 30$ mm, which might be attributed to wavefront distortions.

IX. CONCLUSION AND OUTLOOK

In this paper, we have shown that a low-finesse fiber-optic Fabry-Pérot interferometer is suitable for displacement sensing in a wide range of applications by using different configurations of the sensing cavity. The interferometer enables remote sensing by means of optical fibers and is, therefore, well suited to work in extreme environments like cryogenic temperatures, ultra high vacuum, or high magnetic fields. The possible working ranges and angular alignment tolerances of the presented configurations of the Fabry-Pérot cavity have been characterized experimentally and theoretically with respect to the interference contrast which is a measure for the signal quality. This can give assistance in finding the right sensor head configuration for the desired application. For theoretical characterization of the optical response of the Fabry-Pérot interferometer, a simulation method based on the combination of the Airy formula with the coupling efficiency calculation provided by Zemax was introduced. The good agreement between measurements and simulations demonstrate that this method is well suited to predict the optical response of arbitrary future sensor head configurations. This simplifies the customization of sensor heads as well as the development of new sensor heads as it reduces the time for prototyping and testing.

The investigation of different configurations of the Fabry-Pérot cavity has shown that small size sensor heads are realized by placing a bare fiber close to a high reflective mirror. Best signal contrast is obtained at a distance of about 0.3 mm, with a measurement range in the order of a 100 μ m and an angle tolerance of more than $\pm 2^\circ$. By using a special confocal double pass arrangement comprising a collimator, the measurement range can be expanded to at least 100 mm. The position dependent angular alignment tolerance around the excepted normal incidence angle is ranging from $\pm 1^\circ$ at minimum distance to about $\pm 0.2^\circ$ at maximum distance. This arrangement makes the interferometer easy to use and insensitive to lateral displacements of the moving target. For further extending the measurement range up to 400 mm, a high reflective corner cube can be used as target reflector, but with a reduced contrast of about 0.4 for normal incidence, which is still more than good enough for any practical devices. The use of a focusing lens, moreover, allows the measurement to be made on rough, low reflective, or small targets with mea-

surement ranges in the order from hundreds of μ m to several millimeters, depending on the reflectivity and the degree of beam focusing. High target reflectivity can be compensated by a reduction of the fiber coupling efficiency, realized by defocusing the target.

Future work should evaluate the dependence of the displacement measurement resolution and accuracy on coupling efficiency changes in order to further specify the signal within the possible working ranges. It further should deal with the correction of nonlinearities arising from the Fabry-Pérot cavity.

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- ¹H. Bosse and G. Wilkening, *Meas. Sci. Technol.* **16**, 2155 (2005).
- ²P. Henshaw and S. Lis, in *Proceedings of the Advanced Semiconductor Manufacturing Conference* (New York, 1995), p. 81.
- ³J. E. Griffith and D. A. Grigg, *J. Appl. Phys.* **74**, R83 (1993).
- ⁴W. Estler, K. Edmundson, G. Peggs, and D. Parker, *CIRP Ann.* **51**, 587 (2002).
- ⁵S. Redaelli, in *Proceedings of the CAS – CERN Accelerator School: Course on Beam Diagnostics, Dourdan, France* (Geneva, 2008), p. 535.
- ⁶C. K. Kirkendall and A. Dandridge, *J. Phys. D: Appl. Phys.* **37**, R197 (2004).
- ⁷J. R. Lawall, *J. Opt. Soc. Am. A* **22**, 2786 (2005).
- ⁸M. Durand, J. Lawall, and Y. Wang, *Meas. Sci. Technol.* **22**, 094025 (2011).
- ⁹D. T. Smith, J. R. Pratt, and L. P. Howard, *Rev. Sci. Instrum.* **80**, 035105 (2009).
- ¹⁰D. Rugar, H. J. Mamin, and P. Guethner, *Appl. Phys. Lett.* **55**, 2588 (1989).
- ¹¹M. Born and E. Wolf, *Principles of Optics* (Cambridge University Press, 1999).
- ¹²R. E. Wagner and W. J. Tomlinson, *Appl. Opt.* **21**, 2671 (1982).
- ¹³Y. St-Amant, D. Gariépy, and D. Rancourt, *Appl. Opt.* **43**, 5691 (2004).
- ¹⁴P. Nachman and A. C. Bernstein, *Am. J. Phys.* **65**, 202 (1997).
- ¹⁵A. Dandridge, A. Tveten, and T. Gialllorenzi, *IEEE J. Quantum Electron.* **18**, 1647 (1982).
- ¹⁶K. Karrai, EU patent application EP2045572 (8 April 2009).
- ¹⁷Zemax optical design software, Version Zemax 12 R2 SE, Radiant Zemax, LLC.
- ¹⁸Corning SMF-28 Optical Fiber, Product Information, Corning Inc., 2002.
- ¹⁹K. Karrai and P.-F. Braun, U.S. patent application US 20110211199 A1 (1 September 2011).
- ²⁰S. Feng and H. G. Winful, *Opt. Lett.* **26**, 485 (2001).
- ²¹B. W. Hoogenboom, P. L. T. M. Frederix, J. L. Yang, S. Martin, Y. Pellmont, M. Steinacher, S. Zäch, E. Langenbach, H.-J. Heimbeck, A. Engel, and H. J. Hug, *Appl. Phys. Lett.* **86**, 074101 (2005).
- ²²J. Kim, K. Lauer, H. Yan, Y. S. Chu, and E. Nazaretski, *Rev. Sci. Instrum.* **84**, 035006 (2013).